



Effects of heat stress on close-up dairy cows and newborn heifer calf weights in a humid continental climate

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ABSTRACT

In humid continental climates, dairy cattle are exposed to episodic periods of high temperature and humidity for several months each year. This environment is characterized by large fluctuations in temperature between seasons and even within the same day, creating unique challenges for maintaining thermal balance and animal performance. However, the impact of prenatal heat stress during the close-up period on calf birth weight in this type of climate has not been well established. The objective of this observational study was to determine whether close-up cows exposed to summer conditions of a humid continental climate show physiological signs of heat stress, and whether heifer calves born to cows gestating under humid summer conditions have lower birth weights. Close-up Holstein dairy cows (82 nulliparous, 51 parous) were enrolled at 2 commercial dairy farms in Ontario, Canada, during the close-up period (–21 d to calving) between May and August of 2024. Cow gestation length averaged 276 ± 5 d (mean $\pm$ SD). The farms were visited 3 times per week for the study period, with cow respiration rate, rectal temperature, and skin temperature recorded at each visit for a subset of 102 cows that were bred with sexed semen (71 nulliparous, 31 parous). Heifer calves ($n = 133$) were weighed after birth, with calves averaging 1.1 ± 0.9 d of age at weighing; calves were weighed on the day of calving (d 0; $n = 39$), on d 1 ($n = 57$), and on d 2 to 4 ($n = 37$). Data loggers placed within pens at just above cow level recorded ambient temperature and relative humidity at 10-min intervals. Temperature-humidity indices (THI) were calculated for each cow's close-up period, including average daily THI, maximum THI, and average hours per day for which THI exceeded the thresholds of 62, 65, 68, and 72. Daily THI averaged 67.0 ± 3.25 (range = 59.5–73.1). The daily maximum THI averaged 72.7 ± 3.07 (range =

65.1–78.7). The average (range) time per day above the THI thresholds of 62, 65, 68, and 72 was 19.0 ± 4.19 h/d (7.2–24.0 h/d), 15.3 ± 5.05 h/d (2.7–23.8 h/d), 11.5 ± 5.36 h/d (0.43–23.0 h/d), and 5.3 ± 3.60 h/d (0.0–13.8 h/d), respectively. Cow respiration rate and both average and maximum skin temperatures were positively associated with all THI metrics. Calf birth weight averaged 41.0 ± 4.5 kg (range = 31.3–53.5 kg). For each 1-unit increase in daily average THI and maximum THI, calf birth weight ($\pm$ SE) was reduced by 0.45 ± 0.18 kg and 0.44 ± 0.17 kg, respectively. For each 1-h increase in time (h/d) the THI exceeded 62, 65, 68, and 72, birth weight decreased by 0.24 ± 0.12 kg, 0.30 ± 0.12 kg, 0.29 ± 0.11 kg, and 0.47 ± 0.17 kg, respectively. Overall, the results indicate that close-up cows in moderate, yet humid continental climate summer conditions experience negative consequences, specifically physiological signs of heat stress and reduced calf birth weight; this highlights the need to implement cooling strategies for close-up cows even in temperate regions.

Key words: heat stress, birth weight, close-up, physiological indicators

INTRODUCTION

Dairy cattle maintain a relatively stable core body temperature by balancing the production and loss of heat through thermoregulation, regardless of fluctuations in their external environment (Mount and Monteith, 1974). The thermoneutral zone provides a general ambient temperature range in which no extra energy is required to maintain body temperature; for a dairy cow this can range between 5°C and 25°C and may be exceeded during the summer months in many climate regions (Das et al., 2016; Țogoe and Mincă, 2024). Actual heat load is also affected by various additional environmental factors, such as humidity, air movement, and solar radiation (Dikmen and Hansen, 2009). Heat load should therefore be evaluated using an integrated index such as the temperature-humidity index (THI), which better reflects the combined effect of temperature and humidity. Heat stress

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The list of standard abbreviations for JDS is available at adsa.org/jds-abbreviations-26. Nonstandard abbreviations are available in the Notes.

occurs when the external thermal load induces a physiological response, thus surpassing the upper critical limit of the animal's thermal neutral zone (Das et al., 2016).

Heat stress negatively affects the behavior, physiology, metabolism, health, reproduction, and productivity of dairy cattle in all life stages, resulting in economic losses and compromised animal welfare (St-Pierre et al., 2003; Wheelock et al., 2010). Cows experiencing heat stress have reduced feed intake and greater physiological stress; for lactating dairy cows, this is associated with decreases in milk yield and components (Wheelock et al., 2010; Das et al., 2016). Dry dairy cows exposed to heat stress have reduced milk yield in their subsequent lactation due to stunted mammary involution, whereas their gestating calves experience both pre- and postnatal effects (Karimi et al., 2015; Dado-Senn et al., 2020; Ouellet et al., 2020).

Maternal environments can influence prenatal development, affecting calf birth weight, growth, behavior, and future performance across multiple generations (Laporta et al., 2020; Dallago et al., 2022). Prenatal heat stress can affect calf growth and organ development; calves born to cows that are heat stressed for 45 d before calving in subtropical climates often weigh less at birth and weaning and have reduced passive immunity transfer, decreased total plasma protein, reduced ADG, and impaired cellular immune functions compared with calves born from cows provided heat abatement via fans and sprinklers (Tao et al., 2012) or fans, water soakers, and shade (Dado-Senn et al., 2020). Calves born with a lower birth weight are at a higher risk of morbidity and mortality (Urie et al., 2018). These effects can extend into the preweaning period, such that calves exposed to prenatal heat stress 45 d before calving have a lower survival rate to first calving and produce less milk upon entering the milking herd, which can persist up to 3 lactations, affecting heifer calves' production and longevity in the herd (Monteiro et al., 2016; Dado-Senn et al., 2020; Laporta et al., 2020). In temperate climates, close-up (21 d before calving) heat stress has been demonstrated to affect cross-generation genetic parameters of milk production, as well as health outcomes, including clinical mastitis, endometritis, and digital dermatitis (Kipp et al., 2021). Alternatively, Vinet et al. (2024) assessed heat stress during all stages of gestation for nulliparous and primiparous cattle in France's temperate climate and observed limited effects on their heifer calves' future health and production.

Within North America, some northern parts of the United States and central and southeast regions of Canada have a humid continental climate in which dairy cattle experience episodic periods of high temperatures and humidity for several months per year (Köppen, 1918; Kottek et al., 2006; NOAA, 2023). The humid continental climate can be defined as those areas between latitude 40°N and 60°N of the equator (Köppen, 1918; Kottek

et al., 2006; NOAA, 2023). VanderZaag et al. (2024) performed predictive climate analyses and determined that the Canadian province of Ontario has the highest frequency and duration of heat stress within Canada, which is particularly concerning because Ontario and Quebec are the leading milk-producing provinces in the country. Their analysis also predicted significant increases in THI for this geographic area by 2030, such that levels may reach those similar to current conditions in the southern United States (Key and Sneeringer, 2014). Dairy cattle raised in humid continental climates are not well adapted to sustained heat and humidity, as these cows have been genetically selected for higher milk yield and component production in cooler environments (Misztal et al., 2025). This makes dairy cattle raised in humid continental climates particularly vulnerable to the increasing frequency and duration of heat stress events. The risk of heat stress in regions with a humid continental climate emphasizes the need for further research and mitigation strategies on these farms.

The impact of prenatal heat stress during the close-up period on calf birth weight in this type of climate is limited, and researchers have previously relied on existing databases and retrospective datasets (Kipp et al., 2021; Vinet et al., 2024). Although that approach enables large-scale analyses, it may underestimate actual animal exposure because on-farm microclimatic conditions can differ substantially from those recorded outdoors due to variations in barn design, ventilation, and management practices. In addition, in these retrospective approaches, researchers have typically defined heat stress using universal thresholds, which do not account for individual variation in heat tolerance among animals. Therefore, this study aimed to directly observe the effects of summer conditions on close-up cows on commercial farms in Canada. The objectives were to determine whether summer conditions in a humid continental climate are associated with physiological signs of heat stress in close-up cows, and whether prenatal exposure to these conditions during the close-up period affects heifer calf birth weight. It was predicted that summer conditions would induce physiological signs of heat stress in late-gestating dairy cattle and reduce birth weights in their heifer calves.

MATERIALS AND METHODS

Farm Criteria, Animal Enrollment, and Sample Size

Two commercial dairy farms were recruited and enrolled, both of which met the study criteria of housing Holstein dairy cattle, having a herd size with ~25 to 30 heifer calves born per month, and being located in southwestern Ontario, Canada, within 120 km of the University of Guelph (Guelph, ON, Canada). Pregnant nulliparous

and parous cows bred with conventional and sexed dairy semen were enrolled at close-up (21 d before expected calving) from May to August 2024 and remained in the study until calving. Holstein heifer calves born from these cattle were enrolled at the closest visit after birth (mean $\pm$ SD = 1.1 ± 0.9 d of age; range = 0 to 4 d of age).

Minimum sample sizes were estimated using the POWER procedure of SAS 9.4 (SAS Institute Inc., Cary, NC), considering a power level of 80%, significance at 0.05, and a cumulative effect size of 25% for all fixed effects (aside from the heat stress predictors) considered in the statistical models. Calves born from late gestation cows experiencing heat stress in subtropical climates weigh, on average, $\sim 10\%$ less compared with calves born from cooled cows (Ouellet et al., 2020, 2021a). To conservatively estimate the effect of close-up prenatal heat stress on calf birth weight, an effect size of 5% was selected, which resulted in a calculated minimum sample size of 119 calves; however, more calves were enrolled to account for any potential losses. Tresoldi et al. (2025) determined that in a Mediterranean climate, ambient temperature explained 35% of the variation in respiration rate across cows in all life stages, whereas average daily THI explained 26% of the variation in respiration rate. From this, an effect size of 20% to 40% for heat stress on respiration rate was estimated, which resulted in a calculated minimum sample size ranging from 17 to 32 cows. Because more calves were required, additional cows were enrolled, such that physiological indicators were measured on 102 cows, which corresponded to a calculated power level of 99%.

Animals, Housing, and Management

Farm demographics, housing, and management practices were recorded during farm recruitment before the beginning of data collection. Farms were visited 3 d/wk, with both farms visited on the same day, and the farm visited in the morning was alternated between visits. Measures were in place to ensure proper biosecurity within and between farms.

One farm had a herd size of 450 milking cows and milked by automated milking system (AMS) units (Lely, A5, Maasluis, the Netherlands), including 1 for fresh cows and 10 for lactating cows. There were a series of 4 dynamic close-up pens; nulliparous and parous cows were moved into separate parity pens at 21 d before their expected calving, and then as they approached their expected calving date and other cows calved, they were moved and comingled across parities into another pen at ~ 14 d before calving, and then finally moved into a calving pen at ~ 7 d before calving. These 4 pens were all straw bedded with a target group size of 10 to 15 animals per pen, such that stocking density was >11.1 m²

per cow. Close-up cows were fed a dry cow ration once daily, with feed manually pushed up 4 times per day. Cows had ad libitum access to water from troughs, with 1 trough shared between 2 pens. Fans that were 1.22 m in diameter (MOB4851-2, Schaefer, Sauk Rapids, MN) were mounted at a height of approximately 4 m in close-up pens, averaging 2 fans per pen. The fans were turned on manually at the discretion of the barn staff when an increase in the barn's temperature was observed. Calving was monitored by visual inspection, with cameras monitoring close-up pens. After calving, cows were immediately moved into a fresh cow pen with access to an AMS. Regarding calf care, farm staff bottle-fed newborn calves as soon as possible after birth with a first meal of 4 L of previously frozen colostrum. A second meal of 2 L of colostrum was fed within 12 h of the first feeding. After the second feeding, colostrum was mixed with whole milk and fed to calves in 2.5-L meals 3 times per day for 3 to 5 d. Standard calf care practices within the first week of life included an iodine navel dip, calf bolus (First Defense, Portland, ME), selenium injection (Selon E, Vetoquinol, Lure, France), and an oral dose of Halocur (MSD Animal Health, Merck & Co. Inc., Rahway, NJ). Nursery calves were housed in individual straw-bedded pens near the fresh pen for the first 2 to 4 d of life and were then moved into calf rearing barns.

The second farm had a herd size of 300 milking cows and milked in a parlor twice daily (at 0500 and 1500 h). Nulliparous and parous cows were comingled during the close-up period and until calving and were housed on sand-bedded pack in an open-sided, naturally ventilated barn. The target group size was 35 to 40 animals, such that stocking density was ≥ 10.0 m²/cow. Cattle were fed a close-up ration once daily at the feed bunk and had ad libitum access to water in 2 troughs. Cows may have experienced some sun exposure while at the feed bunk; however, this exposure was not able to be measured, and solar radiation was therefore not assessed. No heat abatement was provided in the close-up pen. Calving was monitored by visual inspection, where cows were moved to a maternity pen within 1 to 2 h after calving. Nursery calves were tube-fed 4 L of pooled colostrum within the first hour of life, or as soon as possible, with a second meal of 3 L of colostrum fed within 12 h of birth. This was followed with 3-L meals of whole milk fed twice per day. Calves were housed in individual straw-bedded pens in the maternity barn for first 2 to 4 d of life, then moved outdoors to individual straw-bedded hutches (Flex-Hutch AGRI-8002, Agri-Plastics, Grassie, ON, Canada) until weaning. Standard calf care within the first week of life included an iodine navel dip and intranasal vaccine (Bovilis Nasalgen 3, Merck & Co. Inc., Rahway, NJ).

A total of 151 calves were born during the study period; this included 3 bull calves, 2 sets of freemartin twins,

6 heifer calves that died before measuring birth weight, and 2 sets of heifer twins. To avoid reduced birth weights associated with twins, as well as freemartins, only heifer calves born as singletons were included. Cows that had bull calves, twins, and heifer calves that died before their first weighing were excluded from analyses, as were 2 cows that died before calving and 2 cows that aborted late term. Therefore, the final calf sample size was 134 heifer calves. Further, a total of 134 close-up cows (83 nulliparous and 51 parous; mean parity of parous cows $\pm$ SD = 2.0 ± 1.11 ; range = 0 to 5) were enrolled in this study. A total of 70 cows (41 nulliparous and 29 parous) were enrolled from farm 1, and a total of 64 cows (42 nulliparous and 22 parous) were enrolled from farm 2. Of these 134 cows, those bred with sexed semen, and therefore expected to produce heifer calves, formed a subset of 102 cows (71 nulliparous and 31 parous), which also had physiological measurements recorded. The average ($\pm$ SD) gestation length was 276 ± 5 d (range = 258 to 290 d).

Experimental Design and Data Collection

On the first farm, 4 temperature-humidity data loggers (Kestrel DROP D2AG Livestock Heat Stress Monitor, Nielsen-Kellerman, Boothwyn, PA) were placed centrally and throughout the 4 close-up pens, with 1 per pen. On the second farm, 2 temperature-humidity data loggers were placed medially and spaced evenly throughout the barn. All loggers were mounted ~2 m above the ground during the first farm visit. These data loggers recorded temperature and relative humidity in 10-min intervals, and data were downloaded once per week. Wind speed (m/s) was recorded at each visit using an anemometer (BT-100-WM, BTMETER, Bestmeter Zhuhai Electronic Technology Ltd., Zhuhai, Guangdong, China) placed at cow level; with the orientation manipulated until the maximum wind speed was located. This was done at each corner of the close-up pens and in 2 positions in the middle of the pens (total 6 positions per pen). Average daily wind speed was calculated for each farm.

Body condition score was recorded for each cow bred with sexed semen at ~21 d before expected calving using a 5-point scale with 0.25-point increments (Wildman et al., 1982). Intra- and interobserver reliability of $\kappa > 0.94$ were established for 2 trained observers before the start of data collection. At each farm visit, respiratory rate, rectal temperature, and rump skin temperature were recorded for each of these cows within 21 d of expected calving. These data were summarized per week relative to calving (−3, −2, −1) by averaging values per day. Respiratory rate was recorded visually before handling; researchers counted flank movements for 10 s, and this value was multiplied by 6 to obtain breaths per minute. Two trained

observers measured respiratory rate; reliability was assessed before and throughout data collection, and due to the obvious nature of respiratory rate, observers were always consistent. Rectal temperature was measured with a digital thermometer with a 12.7-cm probe length (SHARPTemp V Digital Thermometer, Portsmouth, RI) by placing the probe into and against the rectum until the temperature stabilized and the measurement was complete. Rump skin temperature was measured using an infrared camera (Infrared FLIR E8-XT Series Camera, model FLIR-E6390, FLIR, Stockholm, Sweden); researchers captured 2 photos per cow (one of each side of the rump) at each farm visit. A single observer performed thermal imaging analyses on FLIR Thermal Studio (The World's Sixth Sense, Teledyne FLIR LLC 2025, Thousand Oaks, CA). The target area was the circular area under the natural anatomical curvature of the tail. Before starting, interobserver reliability was confirmed against a trained observer, and intra-observer reliability was confirmed before starting as well as midway through to ensure no observational drift (interclass correlation coefficient > 0.98). The mean maximum and average daily skin temperatures were calculated from the 2 rump images per cow per day.

All heifer calves were weighed by researchers at the closest visit after birth (1.0 ± 0.9 d of age; mean $\pm$ SD) using a portable digital animal scale (Caf Cart Digital Small Animal Scale, Raytec LLC, Ephrata, PA). Calves were weighed on the day of calving (d 0; $n = 39$), on d 1 ($n = 57$), on d 2 ($n = 23$), on d 3 ($n = 13$), and on d 4 ($n = 1$). Newborn calves were weighed immediately upon arrival of researchers at the farm for both morning (a.m.) and afternoon (p.m.) visits. The scale was brought to each calf and zeroed before calves were guided into the scale to have their weight recorded.

THI Calculations and Thresholds

The ambient temperature and relative humidity data were used to calculate THI for each 10-min measurement and were thus averaged across loggers on each farm, which corresponded to 144 measurements per day per farm. The THI was calculated for each measurement using the equation developed by the NRC (1971):

$$\text{THI} = (1.8 \times T + 32) - [(0.55 - 0.0055 \times \text{RH}) \times (1.8 \times T - 26)],$$

where T = ambient temperature ($^{\circ}\text{C}$) and RH = relative humidity (%).

A total of 144 THI values were calculated for each day of each animal's close-up period (21 d before calving) to determine the following metrics: average daily THI,

maximum daily THI, and the time (h/d) for which THI exceeded the thresholds of 62, 65, 68, and 72. Traditionally, higher THI thresholds have been used in research to assess heat stress, although in recent literature, heat stress has been reported to occur at even lower THI levels. A THI of 72 has commonly been used as a threshold for lactating dairy cows (Ravagnolo et al., 2000). In continental climates, lower THI values ranging from 50 to 70 have been identified as breakpoints for heat stress indicators and cow performance in lactating cows (Müschner-Siemens et al., 2020; Pinto et al., 2020; Campos et al., 2022). Temperature-humidity index thresholds of 68 and 72 have been applied to assess heat stress in dry cows (Fabris et al., 2019; Stefanska et al., 2024), although these values were largely derived from research on lactating cows and have not been specifically validated for dry cows. Given this uncertainty, and evidence that heat stress may occur at lower THI values in continental climates (Müschner-Siemens et al., 2020; Pinto et al., 2020; Campos et al., 2022), a range of thresholds (62, 65, 68, and 72) were evaluated in this study. Average daily THI was calculated by taking the mean of the 144 measurements per day. The maximum daily THI corresponded to the measurement with the highest calculated THI per day. The numbers of measurements per day above each threshold were summed and divided by 6 to obtain the time (h/d) above each threshold. Each THI metric was summarized on an individual cow basis per week relative to calving (−3, −2, −1) and also by close-up period, by averaging the values per day.

Statistical Analysis

Statistical analyses were performed using SAS 9.4 (SAS Institute Inc., Cary, NC) with cow and calf as the experimental units for all models. Before analyses, all data were screened and verified for normality using the UNIVARIATE procedure in SAS through visual inspection of the data plots and using the Shapiro–Wilk test (with $P > 0.05$ indicating normality). One calf-cow pair was removed from the dataset, as the calf birth weight was beyond 3 SD of the mean. The final sample sizes included 82 nulliparous and 51 parous cows ($n = 133$) with 133 calves, with a subset of 102 cows (71 nulliparous and 31 parous) having physiological measurements recorded.

The MIXED procedure of SAS was used to model the association of various explanatory variables on cow physiological indicators (respiratory rate, average and maximum skin temperature, rectal temperature) and calf birth weight. Significance was denoted at $P \leq 0.05$ and tendencies were declared as $0.05 < P \leq 0.10$ for all models.

The physiological cow indicators of heat stress and THI metrics were averaged per week relative to calving,

and each physiological indicator was modeled separately for each THI metric as a predictor (average daily THI, maximum daily THI, hours per day above THI thresholds of 62, 65, 68, 72). Week relative to calving was included as a fixed effect in a repeated measures model, with cow nested within farm as the subject and compound symmetry as the covariance structure. Compound symmetry was used as this generated the lowest fit statistic for the Akaike information criterion. The model included the random effect of calving month. Parity (nulliparous vs. parous) was also included in each model as a fixed effect to account for parity specific responses, and farm was considered as a fixed effect to account for variation in management between the 2 farms. Wind speed and cow BCS were tested as covariates and retained in the models if $P < 0.1$. An interaction between farm and each THI metric was specifically tested to account for differences in cooling strategies between farms; however, this interaction was not significant ($P > 0.05$) and thus not retained in any of the models. An interaction between parity and THI metric was also tested; this interaction was not significant ($P > 0.05$) and thus not retained in any of the models. For all other variables kept in the models, we examined 2-way interactions, and no interactions ($P > 0.05$) were detected.

Each THI metric was also averaged per close-up period, which considered all 3 wk relative to calving. Calf birth weight was modeled separately using each THI metric as a predictor (average daily THI, maximum daily THI, hours per day above threshold of 62, 65, 68, 72). The model included the random effect of calving month. Parity and gestation length were included in each model as fixed effects. Given the potential effect of heat stress to reduce gestation length (Ouellet et al., 2020, 2021b), we first tested to see if gestation length was associated with THI metrics, but we did not detect any significant relationships. From this, we then used gestation length as a fixed effect to account for its known effect on birth weight. To account for the variation in time of weighing relative to colostrum or milk feeding, the time of weighing (a.m. or p.m.) was included in each model. We also tested for an interaction between gestation length and THI metrics; however, this interaction was not significant and thus was not retained in any of the models. The day of age at calf weighing was also included in each model as a fixed effect to account for variation in age upon measuring birth weight, and farm was included as a fixed effect in each model to account for variation in management between the 2 farms. An interaction between farm and each THI metric was specifically tested in order to account for cooling strategies and other housing and management factors that differed between farms; however, this interaction was not significant ($P > 0.05$) and thus was not retained in any of the models. Interactions

Table 1. Descriptive statistics of THI metrics, cow physiological variables (n = 102), cow BCS (n = 78), and calf birth weight (n = 133)

Variable	Mean	SD	Minimum	Maximum
Average THI ¹	67.0	3.25	59.5	73.1
Maximum THI ²	72.7	3.07	65.1	78.7
THI >62 ³ (h/d)	19.0	4.19	7.2	24.0
THI >65 ⁴ (h/d)	15.3	5.05	2.7	23.8
THI >68 ⁵ (h/d)	11.5	5.36	0.4	23.0
THI >72 ⁶ (h/d)	5.3	3.60	0.0	13.8
Average wind speed ⁷ (m/s)	1.1	0.36	0.5	2.1
Rectal temperature ⁸ (°C)	38.9	0.27	38.1	40.1
Respiratory rate ⁹ (breaths/min)	46.0	8.37	28.0	92.0
Average skin temperature ¹⁰ (°C)	32.6	2.13	24.8	37.3
Maximum skin temperature ¹¹ (°C)	34.4	1.63	28.4	38.2
Gestation length ¹² (d)	275.9	4.76	258.0	290.0
Close-up cow BCS ¹³	3.4	0.40	2.5	4.3
Birth weight ¹⁴ (kg)	41.0	4.50	31.3	53.5
Day of age at weighing ¹⁵ (d)	1.1	1.00	0.0	4.0

¹Average daily THI averaged by week of the close-up period, from -3 to -1 wk relative to calving (WRC).

²Maximum THI averaged by week of the close-up period, from -3 to -1 WRC.

³Time (h/d) THI was greater than 62 averaged by week of the close-up period, from -3 to -1 WRC.

⁴Time (h/d) THI was greater than 65 averaged by week of the close-up period, from -3 to -1 WRC.

⁵Time (h/d) THI was greater than 68 averaged by week of the close-up period, from -3 to -1 WRC.

⁶Time (h/d) THI was greater than 72 averaged by week of the close-up period, from -3 to -1 WRC.

⁷Average daily wind speed in the close-up pen measured from 6 locations thrice per week, averaged by week of the close-up period, from -3 to -1 WRC.

⁸Average rectal temperature measured thrice weekly, averaged by week of the close-up period, from -3 to -1 WRC in 102 cows.

⁹Average respiration rate measured thrice weekly, averaged by week of the close-up period, from -3 to -1 WRC in 102 cows.

¹⁰Average skin temperature measured thrice weekly averaged by week of the close-up period, from -3 to -1 WRC in 102 cows.

¹¹Maximum skin temperature measured thrice weekly, averaged by week of the close-up period, from -3 to -1 WRC 102 cows.

¹²Gestation length (n = 133).

¹³BCS of close-up cows at enrollment (~21 d to expected calving date).

¹⁴First weight recorded at visit following birth (n = 133).

¹⁵Calves' day of age when measuring birth weight.

between time of weighing and farm, as well as day of age at weighing, were tested to account for differences in farm feeding schedules; however, these interactions were not significant ($P > 0.05$) and thus not retained in any of the models. Wind speed and BCS were tested as covariates and retained in the models if $P < 0.1$. We examined 2-way interactions for all variables kept in the models, and no interactions ($P > 0.05$) were detected.

RESULTS

The mean average daily THI; maximum daily THI; time (h/d) the THI exceeded 62, 65, 68, and 72; and wind speed are reported in Table 1. The average cow respiratory rate was 46.0 ± 8.37 breaths/min (ranging from 28.0 to 92.0 breaths/min; Table 1). Average respiratory rate was positively associated with average daily THI (Table 2) and maximum daily THI (Table 3) during the close-up period, whereby every 1-unit increase in average daily

THI and maximum daily THI was associated with an increase in respiration rate of 0.76 ± 0.15 and 0.81 ± 0.15 breaths/min, respectively. Similarly, respiratory rate was also associated with average daily time above THI thresholds of 62 (Table 4), 65 (Table 5), 68 (Table 6), and 72 (Table 7). For each 1-h increase in the time (h/d) THI exceeded 62, 65, 68, and 72, respiratory rate increased by 0.54 ± 0.11 breaths/min (Table 4), 0.49 ± 0.094 breaths/min (Table 5), 0.45 ± 0.09 breaths/min (Table 6), and 0.70 ± 0.14 breaths/min (Table 7), respectively.

The average cow rectal temperature was $38.9^\circ\text{C} \pm 0.27^\circ\text{C}$ (ranging from 38.1°C to 40.1°C ; Table 1). No significant associations were detected between rectal temperature and any of the THI metrics (Tables 2 to 7).

The average cow skin temperature was $32.6^\circ\text{C} \pm 2.13^\circ\text{C}$ (ranging from 24.8°C to 37.3°C ; Table 1). Average skin temperature was positively associated with average daily THI (Table 2) and maximum daily THI (Table 3) during the close-up period, whereby every 1-unit increase in av-

Table 2. Multivariable linear regression models for average daily THI and other fixed effects associated with cow physiological variables: respiratory rate (breaths/min), rectal temperature (°C), and average and maximum skin temperature (°C); n = 102 cows

Variable	Respiratory rate ¹ (breaths/min)			Rectal temperature ² (°C)			Average skin temperature ³ (°C)			Maximum skin temperature ⁴ (°C)		
	β^5	SE	P-value	β^5	SE	P-value	β^5	SE	P-value	β^5	SE	P-value
Intercept	-7.90	9.90	0.51	38.98	0.33	<0.001	4.84	2.03	0.14	13.99	1.68	0.014
Average THI ⁶	0.76	0.15	<0.001	0.0004	0.005	0.94	0.42	0.03	<0.001	0.30	0.03	<0.001
Farm												
1 ⁷	-0.47	1.10	0.66	-0.15	0.038	<0.001	0.57	0.20	0.0054	0.35	0.17	0.038
2 ⁸	Referent	—	—	Referent	—	—	Referent	—	—	Referent	—	—
WRC ⁹												
-3	1.50	1.21	0.22	-0.09	0.03	0.012	-0.20	0.23	0.38	-0.02	0.19	0.92
-2	0.21	1.03	0.84	-0.054	0.03	0.066	-0.13	0.19	0.49	0.05	0.16	0.75
-1	Referent	—	—	Referent	—	—	Referent	—	—	Referent	—	—
Parity ¹⁰												
Nulliparous	4.21	1.17	<0.001	0.012	0.04	0.76	0.48	0.22	0.026	0.29	0.18	0.099
Parous	Referent	—	—	Referent	—	—	Referent	—	—	Referent	—	—
Wind speed ¹¹ (m/s)	—	—	—	—	—	—	-0.48	0.27	0.082	-0.43	0.23	0.060

¹Average respiration rate measured thrice weekly, averaged by week of the close-up period, from -3 to -1 WRC.²Average rectal temperature measured thrice weekly, averaged by week of the close-up period, from -3 to -1 WRC.³Average skin temperature measured thrice weekly, averaged by week of the close-up period, from -3 to -1 WRC.⁴Maximum skin temperature measured thrice weekly, averaged by week of the close-up period, from -3 to -1 WRC.⁵Estimated regression coefficient.⁶Average daily THI averaged by week of the close-up period, from -3 to -1 WRC.⁷Farm housed close-up cows in 4 dynamic comingled straw-bedded pens.⁸Farm housed close-up cows in a sand-bedded, open-sided barn.⁹WRC = week relative to calving. -3: -21 to -15 d relative to calving date. -2: -14 to -8 d relative to calving date. -1: -7 to -1 d relative to calving date.¹⁰Subset of 102 cows: 71 nulliparous and 31 parous.¹¹Average daily wind speed in the close-up pen measured from 6 locations thrice weekly, averaged by week of the close-up period, from -3 to -1 WRC.

Table 3. Multivariable linear regression models for maximum daily THI and other fixed effects associated with cow physiological variables: respiratory rate (breaths/min), rectal temperature (°C), and average and maximum skin temperature (°C); n = 102 cows

Variable	Respiratory rate ¹ (breaths/min)			Rectal temperature ² (°C)			Average skin temperature ³ (°C)			Maximum skin temperature ⁴ (°C)		
	β^5	SE	P-value	β^5	SE	P-value	β^5	SE	P-value	β^5	SE	P-value
Intercept	-15.56	11.06	0.29	38.90	0.36	<0.001	0.72	2.18	0.77	10.80	1.81	0.027
Maximum THI ⁶	0.81	0.15	<0.001	0.0014	0.005	0.78	0.44	0.03	<0.001	0.33	0.02	<0.001
Farm												
1 ⁷	-0.51	1.10	0.64	-0.15	0.04	<0.001	0.58	0.19	0.0035	0.36	0.16	0.028
2 ⁸	Referent	—	—	Referent	—	—	Referent	—	—	Referent	—	—
WRC ⁹												
-3	1.38	1.21	0.26	-0.090	0.03	0.012	-0.25	0.22	0.25	-0.06	0.18	0.75
-2	0.22	1.03	0.83	-0.053	0.03	0.065	-0.13	0.19	0.49	0.05	0.15	0.73
-1	Referent	—	—	Referent	—	—	Referent	—	—	Referent	—	—
Parity ¹⁰												
Nulliparous	4.24	1.17	<0.001	0.013	0.04	0.76	0.48	0.21	0.025	0.29	0.17	0.098
Parous	Referent	—	—	Referent	—	—	Referent	—	—	Referent	—	—
Wind speed ¹¹ (m/s)	—	—	—	—	—	—	-0.60	0.27	0.026	-0.51	0.22	0.021

¹ Average respiration rate measured thrice weekly, averaged by week of the close-up period, from -3 to -1 WRC.² Average rectal temperature measured thrice weekly, averaged by week of the close-up period, from -3 to -1 WRC.³ Average skin temperature measured thrice weekly, averaged by week of the close-up period, from -3 to -1 WRC.⁴ Maximum skin temperature measured thrice weekly, averaged by week of the close-up period, from -3 to -1 wk WRC.⁵ Estimated regression coefficient.⁶ Maximum daily THI averaged by week of the close-up period, from -3 to -1 WRC.⁷ Farm housed close-up cows in 4 dynamic comingled straw-bedded pens.⁸ Farm housed close-up cows in a sand-bedded, open-sided barn.⁹ WRC = week relative to calving. -3: -21 to -15 d relative to calving date. -2: -14 to -8 d relative to calving date. -1: -7 to -1 d relative to calving date.¹⁰ Subset of 102 cows: 71 nulliparous and 31 parous.¹¹ Average daily wind speed in the close-up pen measured from 6 locations thrice weekly, averaged by week of the close-up period, from -3 to -1 WRC.

Table 4. Multivariable linear regression models for time (h/d) THI >62 and other fixed effects associated with cow physiological variables: respiratory rate (breaths/min), rectal temperature (°C), and average and maximum skin temperature (°C); n = 102 cows

Variable	Respiratory rate ¹ (breaths/min)			Rectal temperature ² (°C)			Average skin temperature ³ (°C)			Maximum skin temperature ⁴ (°C)		
	β^5	SE	P-value	β^5	SE	P-value	β^5	SE	P-value	β^5	SE	P-value
Intercept	33.03	2.51	0.0057	39.02	0.10	<0.001	26.47	0.59	<0.001	29.93	0.50	<0.001
THI > 62 ⁶	0.54	0.11	<0.001	-0.0012	0.004	0.80	0.33	0.023	<0.001	0.24	0.02	<0.001
Farm												
1 ⁷	-0.53	1.09	0.63	-0.15	0.04	<0.001	0.60	0.21	0.0046	0.37	0.17	0.033
2 ⁸	Referent	—	—	Referent	—	—	Referent	—	—	Referent	—	—
WRC ⁹												
-3	1.48	1.23	0.23	-0.09	0.03	0.011	-0.16	0.22	0.47	0.01	0.19	0.95
-2	0.35	1.05	0.73	-0.05	0.03	0.066	-0.060	0.19	0.75	0.10	0.16	0.51
-1	Referent	—	—	Referent	—	—	Referent	—	—	Referent	—	—
Parity ¹⁰												
Nulliparous	4.31	1.18	<0.001	0.01	0.04	0.77	0.49	0.22	0.028	0.31	0.18	0.093
Parous	Referent	—	—	Referent	—	—	Referent	—	—	Referent	—	—
Wind speed ¹¹ (m/s)							-0.51	0.27	0.064	-0.47	0.23	0.041

¹Average respiration rate measured thrice weekly, averaged by week of the close-up period, from -3 to -1 WRC.²Average rectal temperature measured thrice weekly, averaged by week of the close-up period, from -3 to -1 WRC.³Average skin temperature measured thrice weekly, averaged by week of the close-up period, from -3 to -1 WRC.⁴Maximum skin temperature measured thrice weekly, averaged by week of the close-up period, from -3 to -1 WRC.⁵Estimated regression coefficient.⁶Time (h/d) THI was greater than 62 averaged by week of the close-up period, from -3 to -1 WRC.⁷Farm housed close-up cows in 4 dynamic comingled straw-bedded pens.⁸Farm housed close-up cows in a sand-bedded, open-sided barn.⁹WRC = week relative to calving. -3: -21 to -15 d relative to calving date. -2: -14 to -8 d relative to calving date. -1: -7 to -1 d relative to calving date.¹⁰Subset of 102 cows: 71 nulliparous and 31 parous.¹¹Average daily wind speed in the close-up pen measured from 6 locations thrice weekly, averaged by week of the close-up period, from -3 to -1 WRC.

Table 5. Multivariable linear regression models for time (h/d) THI >65 and other fixed effects associated with cow physiological variables: respiratory rate (breaths/min), rectal temperature (°C), and average and maximum skin temperature (°C); n = 102 cows

Variable	Respiratory rate ¹ (breaths/min)			Rectal temperature ² (°C)			Average skin temperature ³ (°C)			Maximum skin temperature ⁴ (°C)		
	β^5	SE	P-value	β^5	SE	P-value	β^5	SE	P-value	β^5	SE	P-value
Intercept	35.70	1.94	0.0029	39.00	0.086	<0.001	28.00	0.40	<0.001	30.99	0.33	<0.001
THI > 65 ⁶	0.49	0.094	<0.001	-0.0003	0.003	0.92	0.28	0.019	<0.001	0.20	0.02	<0.001
Farm												
1 ⁷	-0.39	1.10	0.72	-0.15	0.04	<0.001	0.59	0.20	0.0043	0.36	0.17	0.037
2 ⁸	Referent	—	—	Referent	—	—	Referent	—	—	Referent	—	—
WRC ⁹												
-3	1.54	1.22	0.21	-0.09	0.03	0.012	-0.19	0.23	0.42	-0.01	0.19	0.95
-2	0.26	1.03	0.80	-0.03	0.03	0.067	-0.12	0.19	0.52	0.05	0.16	0.74
-1	Referent	—	—	Referent	—	—	Referent	—	—	Referent	—	—
Parity ¹⁰												
Nulliparous	4.29	1.17	<0.001	0.01	0.04	0.77	0.50	0.22	0.024	0.31	0.18	0.089
Parous	Referent	—	—	Referent	—	—	Referent	—	—	Referent	—	—

¹Average respiration rate measured thrice weekly, averaged by week of the close-up period, from -3 to -1 WRC.²Average rectal temperature measured thrice weekly, averaged by week of the close-up period, from -3 to -1 WRC.³Average skin temperature measured thrice weekly, averaged by week of the close-up period, from -3 to -1 WRC.⁴Maximum skin temperature measured thrice weekly, averaged by week of the close-up period, from -3 to -1 WRC.⁵Estimated regression coefficient.⁶Time (h/d) THI was greater than 65, averaged by week of the close-up period, from -3 to -1 WRC.⁷Farm housed close-up cows in 4 dynamic comingled straw-bedded pens.⁸Farm housed close-up cows in a sand-bedded, open-sided barn.⁹WRC = week relative to calving. -3: -21 to -15 d relative to calving date. -2: -14 to -8 d relative to calving date. -1: -7 to -1 d relative to calving date.¹⁰Subset of 102 cows: 71 nulliparous and 31 parous.

Table 6. Multivariable linear regression models for time (h/d) THI >68 and other fixed effects associated with cow physiological variables: respiratory rate (breaths/min), rectal temperature (°C), and average and maximum skin temperature (°C); n = 102 cows

Variable	Respiratory rate ¹ (breaths/min)			Rectal temperature ² (°C)			Average skin temperature ³ (°C)			Maximum skin temperature ⁴ (°C)		
	β ⁵	SE	P-value	β ⁵	SE	P-value	β ⁵	SE	P-value	β ⁵	SE	P-value
Intercept	38.05	1.67	0.0019	38.99	0.07	<0.001	29.34	0.42	<0.001	31.95	0.33	<0.001
THI > 68 ⁶	0.45	0.09	<0.001	0.001	0.003	0.67	0.25	0.02	<0.001	0.18	0.01	<0.001
Farm												
1 ⁷	-0.38	1.11	0.73	-0.15	0.04	<0.001	0.56	0.20	0.0066	0.34	0.17	0.048
2 ⁸	Referent	—	—	Referent	—	—	Referent	—	—	Referent	—	—
WRC ⁹												
-3	1.58	1.23	0.20	-0.09	0.03	0.013	-0.18	0.24	0.44	-0.01	0.20	0.95
-2	0.21	1.04	0.84	-0.05	0.03	0.065	-0.14	0.20	0.48	0.04	0.17	0.81
-1	Referent	—	—	Referent	—	—	Referent	—	—	Referent	—	—
Parity ¹⁰												
Nulliparous	4.18	1.19	<0.001	0.01	0.04	0.75	0.51	0.22	0.022	0.31	0.18	0.085
Parous	Referent	—	—	Referent	—	—	Referent	—	—	Referent	—	—

¹Average respiration rate measured thrice weekly, averaged by week of the close-up period, from -3 to -1 WRC.

²Average rectal temperature measured thrice weekly, averaged by week of the close-up period, from -3 to -1 WRC.

³Average skin temperature measured thrice weekly, averaged by week of the close-up period, from -3 to -1 WRC.

⁴Maximum skin temperature measured thrice weekly, averaged by week of the close-up period, from -3 to -1 WRC.

⁵Estimated regression coefficient.

⁶Time (h/d) THI was greater than 68, averaged by week of the close-up period, from -3 to -1 WRC.

⁷Farm housed close-up cows in 4 dynamic comingled straw-bedded pens.

⁸Farm housed close-up cows in a sand-bedded, open-sided barn.

⁹WRC = week relative to calving. -3: -21 to -15 d relative to calving date. -2: -14 to -8 d relative to calving date. -1: -7 to -1 d relative to calving date.

¹⁰Subset of 102 cows: 71 nulliparous and 31 parous.

Table 7. Multivariable linear regression models for time (h/d) THI >72 and other fixed effects associated with cow physiological variables: respiratory rate (breaths/min), rectal temperature (°C), average and maximum skin temperature (°C); n = 102 cows

Variable	Respiratory rate ¹ (breaths/min)			Rectal temperature ² (°C)			Average skin temperature ³ (°C)			Maximum skin temperature ⁴ (°C)		
	β ⁵	SE	P-value	β ⁵	SE	P-value	β ⁵	SE	P-value	β ⁵	SE	P-value
Intercept	39.82	1.45	0.0013	39.01	0.07	<0.001	31.22	0.57	<0.001	33.40	0.43	<0.001
THI > 72 ⁶	0.70	0.14	<0.001	-0.006	0.004	0.87	0.34	0.03	<0.001	0.25	0.02	<0.001
Farm												
1 ⁷	-0.69	1.11	0.53	-0.15	0.03	<0.001	0.50	0.21	0.016	0.30	0.17	0.075
2 ⁸	Referent	—	—	Referent	—	—	Referent	—	—	Referent	—	—
WRC ⁹												
-3	1.19	1.21	0.33	-0.08	0.03	0.012	-0.35	0.24	0.14	-0.14	0.20	0.49
-2	0.028	1.03	0.98	-0.05	0.03	0.069	-0.20	0.20	0.32	-0.0033	0.17	0.98
-1	Referent	—	—	Referent	—	—	Referent	—	—	Referent	—	—
Parity ¹⁰												
Nulliparous	4.18	1.20	<0.001	0.04	0.04	0.77	0.48	0.22	0.031	0.29	0.18	0.11
Parous	Referent	—	—	Referent	—	—	Referent	—	—	Referent	—	—
Wind speed ¹¹ (m/s)	—	—	—	—	—	—	-0.70	0.29	0.017	-0.60	0.24	0.012

¹Average respiration rate measured thrice weekly, averaged by week of the close-up period, from -3 to -1 WRC.

²Average rectal temperature measured thrice weekly, averaged by week of the close-up period, from -3 to -1 WRC.

³Average skin temperature measured thrice weekly, averaged by week of the close-up period, from -3 to -1 WRC.

⁴Maximum skin temperature measured thrice weekly, averaged by week of the close-up period, from -3 to -1 WRC.

⁵Estimated regression coefficient.

⁶Time (h/d) THI was greater than 72, averaged by week of the close-up period, from -3 to -1 WRC.

⁷Farm housed close-up cows in 4 dynamic comingled straw-bedded pens.

⁸Farm housed close-up cows in a sand-bedded, open-sided barn.

⁹WRC = week relative to calving. -3: -21 to -15 d relative to calving date. -2: -14 to -8 d relative to calving date. -1: -7 to -1 d relative to calving date.

¹⁰Subset of 102 cows: 71 nulliparous and 31 parous.

¹¹Average daily wind speed in the close-up pen measured from 6 locations thrice weekly, averaged by week of the close-up period, from -3 to -1 WRC.

Table 8. Multivariable linear regression models for average daily THI and maximum daily THI and other fixed effects associated with newborn calf weight (n = 133 calves)

Variable	Newborn calf weight (kg)					
	Average THI ¹			Maximum THI ²		
	β^3	SE	P-value	β^3	SE	P-value
Intercept	-69.13	19.81	0.073	-66.39	20.04	0.080
THI metric	-0.44	0.18	0.016	-0.43	0.17	0.012
Farm						
1 ⁴	-2.34	0.59	<0.001	-2.34	0.59	<0.001
2 ⁵	Referent	—	—	Referent	—	—
Parity ⁶						
Nulliparous	-2.19	0.60	<0.001	-2.20	0.60	<0.001
Parous	Referent	—	—	Referent	—	—
Gestation length (d)	0.51	0.06	<0.001	0.51	0.06	<0.001
DOA at weighing ⁷	0.35	0.31	0.26	0.34	0.31	0.27
Time of weighing ⁸						
Morning	-1.04	0.60	0.083	-1.05	0.60	0.080
Afternoon	Referent	—	—	Referent	—	—

¹Average daily THI averaged by each cow's close-up period (-21 to -1 d relative to calving).²Maximum daily THI averaged by each cow's close-up period (-21 d to -1 d relative to calving).³Estimated regression coefficient.⁴Farm housed close-up cows in 4 dynamic comingled straw-bedded pens.⁵Farm housed close-up cows in a sand-bedded, open-sided barn.⁶Sample size of 133 close-up cows (82 nulliparous and 51 parous).⁷Calves' day of age (DOA) when measuring weight.⁸Timing of weighing, related to farm visitation schedule, on the day calf weight was measured.

average daily THI and maximum daily THI was associated with an increase in average skin temperature of $0.42^\circ\text{C} \pm 0.03^\circ\text{C}$ and $0.44^\circ\text{C} \pm 0.03^\circ\text{C}$, respectively. Similarly, average cow skin temperature was also associated with average daily time above THI thresholds of 62 (Table 4), 65 (Table 5), 68 (Table 6), and 72 (Table 7). For each 1-h increase in time (h/d) THI exceeded 62, 65, 68, and 72, average skin temperature increased by $0.33^\circ\text{C} \pm 0.023^\circ\text{C}$ (Table 4), $0.28^\circ\text{C} \pm 0.019^\circ\text{C}$ (Table 5), $0.25^\circ\text{C} \pm 0.02^\circ\text{C}$ (Table 6), and $0.34^\circ\text{C} \pm 0.03^\circ\text{C}$ (Table 7), respectively.

The maximum cow skin temperature was $34.3^\circ\text{C} \pm 1.63^\circ\text{C}$ (ranging from 28.4°C to 38.2°C ; Table 1). Maximum skin temperature was positively associated with average daily THI (Table 2) and maximum daily THI (Table 3) during the close-up period, whereby every 1-unit increase in average daily THI and maximum daily THI was associated with a $0.30^\circ\text{C} \pm 0.03^\circ\text{C}$ and $0.33^\circ\text{C} \pm 0.02^\circ\text{C}$, respective, increase in maximum skin temperature. Similarly, maximum skin temperature was also associated with average daily time above THI thresholds of 62 (Table 4), 65 (Table 5), 68 (Table 6), and 72 (Table 7). For each 1-h increase in time (h/d) THI exceeded 62, 65, 68, and 72, maximum skin temperature increased by $0.24^\circ\text{C} \pm 0.02^\circ\text{C}$ (Table 4), $0.20^\circ\text{C} \pm 0.02^\circ\text{C}$ (Table 5), $0.18^\circ\text{C} \pm 0.01^\circ\text{C}$ (Table 6), and $0.25^\circ\text{C} \pm 0.02^\circ\text{C}$ (Table 7), respectively.

The average calf birth weight was (mean $\pm$ SD) 41.0 ± 4.49 kg (ranging from 31.3 to 53.5 kg; Table 1). Calves

weighed on the day of calving (d 0; n = 39) averaged 41.0 ± 4.62 kg (ranging from 31.3 to 50.3 kg), calves weighed on d 1 (n = 57) averaged 40.7 ± 4.06 kg (ranging from 33.6 to 50.8 kg), and calves weighed on d 2 to 4 (n = 37) averaged 41.6 ± 5.08 kg (ranging from 32.2 to 53.5 kg). Average birth weight was negatively associated with average daily THI and maximum daily THI during the close-up period (Table 8), whereby for every 1-unit increase in average daily THI and maximum daily THI, birth weight decreased by 0.44 ± 0.18 kg and 0.43 ± 0.17 kg, respectively. Similarly, birth weight was also negatively associated with average daily time above THI thresholds of 62, 65, 68, and 72 (Table 9). For each 1-h increase in time (h/d) THI exceeded 62, 65, 68, and 72, birth weight decreased by 0.24 ± 0.12 kg, 0.29 ± 0.11 kg, 0.27 ± 0.11 kg, and 0.45 ± 0.17 kg, respectively.

DISCUSSION

In this observational study, we investigated how varying intensity and duration in the thermal environment during the close-up period affect physiological indicators of heat stress in dairy cows, and whether prenatal exposure to such conditions influences heifer calf birth weight. Humid continental climates are characterized by warm to hot and humid summers, few dry seasons, severe winters, and significant temperature differences between all 4 seasons (NOAA, 2023). Temperature-humidity index is

Table 9. Multivariable linear regression models for time (h/d) THI >62, 65, 68, and 72 and other fixed effects associated with newborn calf weight (n = 133 calves)

Variable	Newborn calf weight (kg)											
	-Hours per day THI > 62 ¹				Hours per day THI > 65 ²				Hours per day THI > 68 ³			
	β^5	SE	P-value		β^5	SE	P-value		β^5	SE	P-value	
Intercept	-92.83	17.09	0.032		-93.50	16.90	0.031		-95.07	16.95	0.030	
THI metric	-0.24	0.12	0.039		-0.29	0.11	0.011		-0.27	0.11	0.016	
Farm												
1 ⁶	-2.28	0.59	<0.001		-2.42	0.60	<0.001		-2.40	0.60	<0.001	
2 ⁷	Referent	—	—		Referent	—	—		Referent	—	—	
Parity ⁸												
Nulliparous	-2.18	0.60	<0.001		-2.20	0.59	<0.001		-2.20	0.60	<0.001	
Parous	Referent	—	—		Referent	—	—		Referent	—	—	
Gestation length (d)	0.51	0.06	<0.001		0.51	0.06	<0.001		0.51	0.06	<0.001	
DOA at weighing (d) ⁹	0.35	0.32	0.27		0.36	0.31	0.26		0.36	0.31	0.25	
Time of weighing ¹⁰												
Morning	-1.09	0.60	0.074		-1.06	0.59	0.078		-1.01	0.60	0.091	
Afternoon	Referent	—	—		Referent	—	—		Referent	—	—	

¹Time (h/d) THI was greater than 62 averaged by each cow's close-up period (-21 to -1 d relative to calving).²Time (h/d) THI was greater than 65 averaged by each cow's close-up period (-21 to -1 d relative to calving).³Time (h/d) THI was greater than 68 averaged by each cow's close-up period (-21 to -1 d relative to calving).⁴Time (h/d) THI was greater than 72 averaged by each cow's close-up period (-21 to -1 d relative to calving).⁵Estimated regression coefficient.⁶Farm housed close-up cows in 4 dynamic comingled straw-bedded pens.⁷Farm housed close-up cows in a sand-bedded, open-sided barn.⁸Sample size of 133 close-up cows (82 nulliparous and 51 parous).⁹Calves' day of age (DOA) when measuring weight.¹⁰Time of weighing, related to farm visitation schedule, on the day calf weight was measured.

commonly used to assess heat stress because it accounts for the effects of both ambient temperature and relative humidity and is effective in evaluating the environmental conditions that dairy cattle experience (West, 2003). In the current study, the mean average daily THI and maximum daily THI for close-up periods between May and August were 67.0 and 72.7, respectively. Campos et al. (2022) analyzed data from weather stations located near Canadian dairy farms in the provinces of Ontario and Quebec from 2010 to 2019 and determined similar environmental conditions between the months of June and September, finding the average daily and maximum daily THI to be 64.6 and 70.0 for Ontario and 61.9 and 67.9, for Quebec, respectively. Ouellet et al. (2019) reported a mean daily THI maximum from June to August of 74.5 in southwest Quebec and 69.2 in eastern Quebec. Similarly, Nordlund et al. (2019) reported a mean daily THI range of 68.5 to 79.0 in a naturally ventilated barn during a heat wave in July 2013 in northeastern Wisconsin, which also has a humid continental climate. THI thresholds for heat stress in lactating cows have been identified through changes in physiological indicators and production levels; these THI thresholds commonly range from 50 to 72 (Ouellet et al., 2019; Pinto et al., 2020; Campos et al., 2022). However, these thresholds are not universal and can vary depending on several factors, including physiological state, milk production level, breed, acclimation history, and the prevailing climate. For example, cows in early lactation or high-producing animals are generally more sensitive to heat load due to higher metabolic heat production, whereas animals adapted to warmer climates may tolerate higher THI values before exhibiting signs of heat stress (Polsky and von Keyserlingk, 2017). In the present study, the THI values observed were representative of summer thermal conditions in a humid continental climate, and although specific thresholds for heat stress in close-up cows under such conditions remain undefined, it is likely that these temperatures and humidity levels contributed to a variable maternal thermal environment and were positively associated with thermoregulatory adjustments during the close-up period.

Increased cow respiration rate can be an early indicator of heat stress, as an attempt to dissipate heat and thermoregulate (Das et al., 2016). We detected a positive association between respiration rate and all THI metrics, such that the modeled increase in respiratory rate ranged from 0.45 to 0.81 breaths/min for each unit increase in THI metric (1 THI unit or 1 h/d above each threshold). This was expected, given that respiration rates increase with increasing ambient temperature and relative humidity (Silanikove, 2000; Brown-Brandl et al., 2005; Yan et al., 2021). Although this detected association was expected, the observed change in respiration rate was modest compared with previously reported values for dry

cows, though direct comparisons with existing literature are limited due to differences in study designs. Our observational study quantified a change in respiration rate per unit of THI; however, in the existing literature (e.g., Toledo et al., 2020; Ouellet et al., 2021b), cows exposed to heat stress with no abatement have been compared with cooled counterparts, and therefore the determined differences were between the cooled and heat stressed groups of cows. Ouellet et al. (2021b) analyzed records of dry cows that experienced THI conditions averaging 76.5, with or without cooling mitigation. Those researchers observed that in Florida's subtropical climate, heat stress increased respiratory rates in dry cows to 72 breaths/min, compared with cooled counterparts, which had an average of 49 breaths/min. Similarly, Toledo et al. (2020) reviewed 6 studies conducted between 2007 and 2014 in Florida. Those researchers determined heat stress conditions increased dry cow respiratory rates to 74 breaths/min, compared with cooled dry cows (48 breaths/min). Comparatively, cows in the present study exposed to the most severe average daily THI conditions (THI = 73.1) had a modeled increase of 10.3 breaths/min compared with cows exposed to the least severe average daily THI conditions (THI = 59.5). In addition, cows in the present study exposed to highest maximum daily THI conditions had a modeled increase of 11 breaths/min relative to cows exposed to lowest maximum daily THI conditions. Although Ouellet et al. (2021b) observed higher average respiratory rates compared with the present study (average = 46 breaths/min), cows in their study were also exposed to more severe heat stress (daily THI = 76.5 ± 0.16) than in the present study, and those researchers identified a THI breakpoint of 77 for an abrupt increase in respiratory rate.

The positive associations between all THI metrics and average and maximum skin temperature were expected, as thermal stress is known to divert blood flow from internal organs and toward the skin to better dissipate heat (Becker et al., 2020). Peng et al. (2019) investigated the effects of an averaged daytime high THI (>78), a moderate THI ($72 \leq \text{THI} \leq 78$), and a low THI (<72) on lactating dairy cows in Beijing, China, which has a hot humid continental climate. In that study, both the mean and maximum rump skin temperatures were highest comparatively when cows were exposed to a high THI, followed by a moderate THI, and then a low THI. Further, those researchers reported the mean and maximum rump skin temperatures at a low THI of 33.95°C and 34.96°C, respectively. These results are consistent with the current study, for which the mean average and maximum skin temperatures were 32.6°C and 34.3°C, respectively. Peng et al. (2019) detected a positive linear relationship between rump skin temperature and THI. Although those authors observed more severe heat stress compared with

the current study, the association of higher skin temperatures with increases in THI metrics they observed even at lower levels indicates a risk of heat stress. Yan et al. (2021) measured skin temperature through thermal images from the side of lactating dairy cows and reported linear increases in skin temperature as THI rose from 68 to 72, which are similar conditions to those recorded in the present study. Researchers have observed similar associations of heat stress with increases in skin temperature of other areas on the body, such as the eye (Shu et al., 2021; Yan et al., 2021), flank, forehead, ear, and udder (Kaufman et al., 2018; Peng et al., 2019); thus supporting the suitability of skin temperature to indicate heat stress.

No associations were detected between cow rectal temperature and any of the THI metrics. This was unexpected, but it may reflect that heat stress conditions in the current study were not particularly severe (with average THI = 67.0), as rectal temperature may lack sensitivity under mild to moderate heat stress (Shu et al., 2021). An additional explanation may be that cows were able to thermoregulate to dissipate heat before core body temperature was affected. Specifically, they may have done this through increased respiration rate, peripheral vasodilation, and other mechanisms not monitored in this study, such as reduced DMI to decrease metabolic heat production or other behavioral adaptations, including reduced lying time (Polsky and von Keyserlingk, 2017). It has been established that dairy cattle exhibit an increase in skin temperature and respiration rate as coping mechanisms before detectable increases in rectal temperature (Silanikove, 2000; Pontiggia et al., 2024). It was also expected that rectal temperature may have been affected by an interaction between farm and THI metrics due to the differences in cooling strategies between the 2 farms; however, no such interactions were detected. Peng et al. (2019) reported rapid changes in rectal temperature of lactating cows that occurred when THI was greater than 74.1, such that at a THI of 81.8, rectal temperature averaged to 39.12°C. Dry cows are more thermotolerant compared with lactating cows, as lactating animals generate more heat due to greater ruminal fermentation resulting from increased DMI (Das et al., 2016; Ouellet et al., 2021b). In a subtropical climate, Ouellet et al. (2021b) determined that dry cows were able to maintain a fairly constant rectal temperature below a THI of 77. In that study, when THI surpassed this threshold, rectal temperature was increased, which indicates that the dry cows in the current study did not experience heat stress to this severity, as the THI only exceeded 72 for an average of 5.3 h/d, (range = 0.0–13.8 h/d), and the highest maximum daily average THI was 78.7. This further supports the conclusion that the cows in the current study experienced mild heat stress, which they were able to cope with by increasing cooling mechanisms through increased respi-

ration and blood flow to the skin, which ultimately led to no detectable differences in rectal temperature.

Although wind speed was retained in the skin temperature models, no associations were detected between wind speed and respiration rate, rectal temperature, or calf birth weight. This was unexpected due to the known effect of air movement to alleviate the environmental thermal load (Yan et al., 2021). Similarly, for all outcomes, we did not detect an interaction between wind speed and farm, which was expected due to the differences in heat abatement between farms. Perhaps the regions of the pen in which wind speed were recorded or the anemometer orientation manipulation were not ideal, and lacked sensitivity. In addition, little variation was observed in the recorded wind speed measurements, which suggests relatively consistent air movement throughout the study period.

Given the recorded THI metrics and associated indicators of heat stress in the close-up cows, we can conclude that the heifer calves of those cows were exposed to some level of prenatal heat stress during the close-up period. Ouellet et al. (2020, 2021b) discussed the reductions in both birth weight and gestation length associated with heat stress. Heat stress directly impairs fetal development through mechanisms such as reduced placental blood flow and altered nutrient partitioning. In addition, fetal growth is also affected through indirect pathways, including reduced gestation length. Although gestation length was not directly associated with THI metrics in our dataset, we accounted for differences in birth weights due to varied gestation lengths in our models to isolate and account for the variability in birth weight associated with this direct effect. This demonstrates that the impact of heat stress on fetal growth occurred even after accounting for earlier calving. Researchers have observed reduced birth weight when calves are exposed to prenatal heat stress for ~45 d before calving in subtropical climates (Tao and Dahl, 2013; Monteiro et al., 2014; Laporta et al., 2017), and in the present study we similarly determined that reduced calf birth weight was associated with higher levels of all THI metrics. Modeled differences of 2.9 kg and 5.9 kg in birth weight were predicted when considering the variation in average daily THI of ± 1 SD (63.8 to 70.3) and ± 2 SD (60.5 to 73.5) from the mean, respectively. These results align with the reduction of 6 kg in birth weight reported by Tao et al. (2012) when calves were exposed to prenatal heat stress for 45 d before calving without mitigation in a humid subtropical climate. Similarly, Karimi et al. (2015) determined a reduction of 2.7 kg in birth weight when calves were exposed to prenatal heat stress in a temperate climate, with an average THI of 69.7. Dado-Senn et al. (2021) reported a reduction of 4 kg in calf birth weight when calves were exposed in utero to THI values (75–83) during the dry period that

were higher than those of the present study, without active cooling. However, given that our model predicted a greater reduction in birth weight when comparing the extremes of average daily THI, this may suggest larger ranges of THI are associated with greater differences in birth weight. This effect of decreased birth weight with prenatal exposure to heat stress was discussed in a review by Ouellet et al. (2020), with an average reduction of 4.6 kg determined from 12 publications, including Tao et al. (2012), where cows were heat stressed in late gestation relative to cooled counterparts in subtropical and Mediterranean climates. Most research on the effects of prenatal heat stress on calf birth weight has been conducted in subtropical and tropical regions. Although some researchers have examined these effects in temperate climates, their analyses were generally based on retrospective datasets matched with outdoor temperature records rather than field studies specifically designed to assess on-farm conditions. Therefore, the present study provides direct evidence from a humid continental climate. Without the provision of heat abatement in humid continental climates during the close-up period, cows experience physiological signs of heat stress and calves are at risk for lower birth weights, as well as postnatal implications such as reduced weaning weights and decreased performance upon joining the milking herd (Tao et al., 2011; Karimi et al., 2015; Monteiro et al., 2016). Thus, this research highlights the importance of mitigating heat stress for close-up cows and supports the need for further research to compare mitigation strategies in humid continental climate. Further research should also explore additional physiological indicators of heat stress such as lying behavior, feed intake, and other internal temperatures (e.g., vaginal), as well as additional cattle breeds, management systems, and climatic regions, in order to ensure the results are generalizable. Finally, calf growth, behavior, and productivity should be observed for long-term effects of prenatal heat stress during the close-up period.

This study is not without limitations. The use of 2 commercial farms for this observational study, though controlled for statistically, introduced variation due to differences between animal management, housing, heat abatement, nutrition, and farm protocols, which may have affected calf birth weight. Specifically, the differences in colostrum feeding and early life calf management should be considered, as well as the variation in time of day of weighing, and calf age at weighing. Researchers were not present on the day of each calving, and due to the commercial setting, some relevant calf records were limited. Researchers were not able to record timing or exact volume of colostrum or milk fed in relation to weighing calves, which may have contributed to inflated birth weight measurements, especially given the variation in

day of age at weighing. To best account for some of this variation, age at weighing, time of weighing, and farm were controlled for in the statistical models.

CONCLUSIONS

Summer conditions in a humid continental climate were positively associated with physiological responses indicative of heat stress in close-up dairy cows. Specifically, close-up cows experienced increases in respiratory rate, and average and maximum skin temperature, with greater daily average THI, daily maximum THI, and time per day above THI thresholds of 62, 65, 68, and 72. Exposure to prenatal heat stress during the close-up period was associated reduced heifer calf birth weight. Overall, this work is novel for a humid continental climate and highlights the importance of mitigating heat stress in the close-up period for the health and comfort of dry cows, the prenatal growth of heifer calves, and the long-term performance and success of both cows and calves.

NOTES

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Nonstandard abbreviations used: AMS = automated milking system; DOA = day of age; THI = temperature-humidity index; WRC = week relative to calving.

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